

erogenous, multifaceted, and distributed sensors talk to each other to create a sensor mesh for solving complicated problems.

Sensor to cloud (S2C) integration. Cyber physical systems (CPS) emerge at the intersection of physical and virtual/cyber worlds.

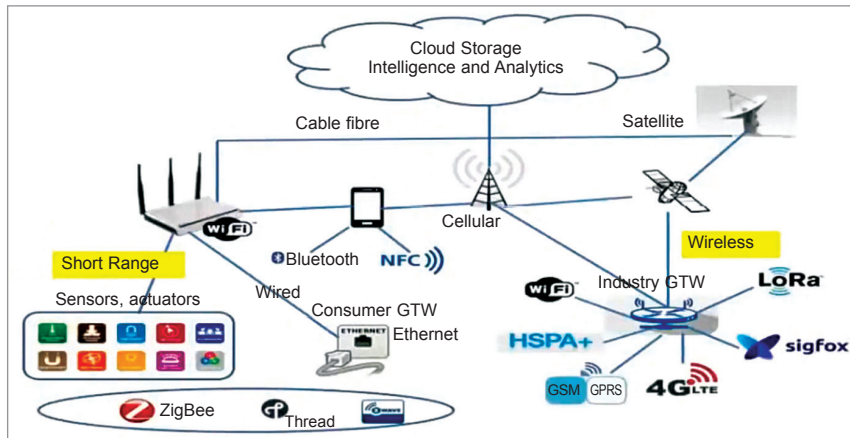


Fig. 2: Network architecture

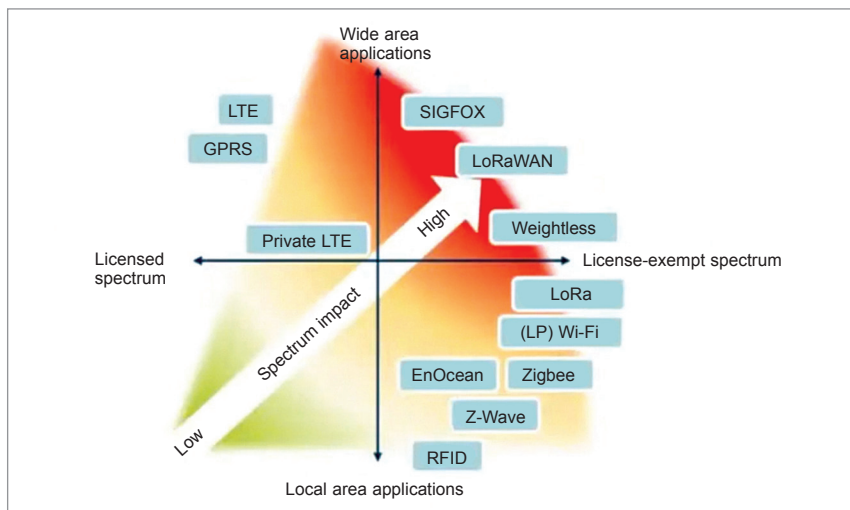


Fig. 3: Wireless network comparisons

Carrier frequency		Technology	Channel bandwidth	Representative data rate	Link budget target or max. range
Licensed cellular		LTE Cat. 0	20MHz	DL: 1Mb/s UL: 1Mb/s	140dB
		LTE Cat. M	1.4MHz	DL: 1Mb/s UL: 1Mb/s	155dB
		NB-IoT	200kHz	DL: 128kb/s UL: 64kb/s	164dB
		EC-GSM	200kHz	DL: 74kb/s UL: 74kb/s	164dB
Unlicensed	2.4GHz	Ingenu RPMA	1MHz	UL: 624kb/s DL: 156kb/s	500km line of sight
	Sub-1 GHz	LoRa chirp spread spectrum	125kHz	UL: 100kb/s DL: 100kb/s	15km rural 5km urban
	Sub-1 GHz	Weightless-N	200Hz	UL: 100b/s	3km urban
	Sub-1 GHz	Sigfox	160Hz	UL: 100b/s	50km rural 10km urban

Fig. 4: IoT Long Range technical solutions

Device to device (D2D) integration.

Device to enterprise (D2E) integration. In order to have remote and real-time monitoring, management, repair, and maintenance, and for enabling decision-support and expert systems, ground-level heterogeneous devices have to be synchronised with control-level enterprise packages such as ERP, SCM, CRM, and KM.

Device to cloud (D2C) integration.

Cloud to cloud (C2C) integration.

Mobile edge computing (MEC), cloudlets, and edge cloud formation through the clustering of heterogeneous edge/fog devices.

When we talk about IoT, we generally refer to connectivity among millions and billions of devices. However, it is difficult to interface them all.

As can be seen, networks such as short-range, wireless, cellular, satellite, and fibre optics are available. For IoT, we have to see how feasible the connectivity is for end devices. Generally, it has to be low-power and low-data type to connect to multilayer devices. GSM seems to fit into this requirement.

Network connectivity issues

- Sensors cannot be easily connected to the power source.
- Sensors deployed in secure or hostile operating conditions are difficult to change without impacting the system.
- Sensors have very different life-time expectancy rate.
- Resource-constrained endpoints.
- High sensitivity on reception.

Some possible options available to us are:

1. Direct connectivity

- Easiest to install and configure.
- Combines machine logic with IoT logic.

However, it is complex, not possible for legacy systems, and is less secure.

2. Cloud gateways

- Devices come with embedded connectivity to a proprietary cloud.
- Forwarding capabilities available to IoT platforms.
- Easy to start in the cloud.

3. Field gateways

- Enables legacy systems.
 - More secure.
- It requires extra hardware, machine logic is separated from IoT logic, and